# Curd Incubator

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

**Covers these lab items:** Small Incubation Box/Curd Maker

## What it is used for

A curd incubator or small curd-making box maintains a controlled temperature that allows a suitable starter culture to ferment milk into curd. Hygienic preparation, accurate temperature control, and adequate incubation time help produce safe, consistent curd.

## Main parts

- Insulated incubation chamber
- Heating element or temperature-control system
- Thermostat or digital temperature controller
- Temperature sensor or display
- Shelves, trays, or container supports
- Power cord, switch, and protective housing

## Before you start

1. Confirm that the milk has been properly heat-treated and cooled to the starter culture's recommended inoculation temperature.
2. Use clean, sanitised food-grade containers and utensils; wash hands and follow the dairy hygiene procedure.
3. Check that the incubator chamber, shelves, door seal, sensor, and power cable are clean and undamaged.
4. Verify that the temperature controller and sensor are functioning and that the chamber can maintain the required temperature.
5. Prepare the starter culture according to its instructions and check its suitability and expiry date.
6. Place the incubator on a stable, dry, ventilated surface away from direct sunlight and water splashes.

## How to use

1. Switch on the incubator and set the temperature recommended for the starter culture and curd process.
2. Allow the chamber to stabilise and verify the displayed temperature before loading the containers.
3. Add the recommended quantity of starter culture to the prepared milk and mix gently using a sanitised utensil.
4. Fill clean containers to a suitable level and cover them to protect against contamination.
5. Arrange containers evenly inside the chamber, leaving space for air circulation where the design requires it.

6. Close the door and start the incubation period according to the starter culture and process instructions.
7. Monitor the chamber temperature at the specified intervals and record the start time and temperature readings.
8. Avoid unnecessary door opening, shaking, or moving the containers during fermentation.
9. Check curd setting after the recommended incubation period using the approved quality check.
10. Stop incubation when the required texture and acidity are reached; do not rely on time alone if the process requires product checks.
11. Remove containers with clean hands or suitable clean handling tools and close them promptly.
12. Cool the curd promptly in a suitable refrigerator to slow further fermentation and maintain quality.
13. Switch off and clean the chamber and removable supports after use, following the manufacturer's instructions.

## ⚠ Safety

- Use only food-grade containers and utensils and maintain strict dairy hygiene.
- Do not handle electrical switches or plugs with wet hands.
- Keep the incubator away from water, wet floors, and sources of contamination.
- Do not exceed the equipment's specified temperature range or block ventilation openings.
- Use care when handling warm milk and hot surfaces to avoid burns.
- Refrigerate finished curd promptly and discard product when contamination or unsafe processing is suspected.

## Care & storage

- Clean and sanitise the chamber, shelves, and container supports after use according to the manufacturer's guidance.
- Check the door seal, power cable, temperature sensor, and controller regularly.
- Verify temperature accuracy periodically using a suitable calibrated thermometer.
- Keep ventilation openings clear and avoid pouring water directly onto electrical components.
- Arrange servicing if temperature fluctuates, heating fails, or the controller gives unreliable readings.

## Common mistakes to avoid

- Adding starter culture when milk is too hot or too cold for the culture.
- Using unclean containers or utensils that can contaminate the milk.
- Setting an incorrect incubation temperature or opening the chamber repeatedly.
- Leaving finished curd warm for too long instead of cooling it promptly.

*Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.*